

Proof. **Theorem (Triangle Inequality).** Let $(V, \langle \cdot, \cdot \rangle)$ be a real inner-product space and denote the induced norm by $\|v\| = \sqrt{\langle v, v \rangle}$. For all vectors $x, y \in V$ one has

$$\|x + y\| \leq \|x\| + \|y\|.$$

1. Degenerate case. If $x = 0$ or $y = 0$ then $\|x + y\| = \|x\| + \|y\|$ because $\|0\| = 0$. Hence the inequality holds with equality.

2. Squaring the inequality. Assume $x \neq 0$ and $y \neq 0$. Both sides are non-negative, so we may square:

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (1)$$

3. Expansion of the squared norm. Using bilinearity and symmetry,

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2. \end{aligned} \quad (2)$$

(The identity was verified with `verify_arithmetic`.)

4. Cauchy–Schwarz bound for the mixed term. The Cauchy–Schwarz inequality gives

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Hence

$$\langle x, y \rangle \leq \|x\| \|y\|. \quad (3)$$

Multiplying by 2,

$$2\langle x, y \rangle \leq 2\|x\| \|y\|. \quad (4)$$

5. Substituting the bound. Insert (4) into (2):

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2. \end{aligned} \quad (5)$$

Thus (5) establishes (1).

6. Taking square roots. Both sides of (1) are non-negative, so taking square roots preserves the inequality:

$$\|x + y\| \leq \|x\| + \|y\|.$$

7. Equality condition. Equality can occur only when equality holds in the Cauchy–Schwarz step (3). Cauchy–Schwarz is an equality iff x and y are linearly

dependent, i.e. $y = \lambda x$ for some $\lambda \in \mathbb{R}$. From (4) we need $2\langle x, y \rangle = 2\|x\| \|y\|$, which forces $\lambda \geq 0$. Thus

$$\|x + y\| = \|x\| + \|y\| \iff y = \lambda x \text{ with } \lambda \geq 0$$

(and symmetrically $x = \mu y$ with $\mu \geq 0$), including the degenerate case when one vector is zero.

Conclusion. All cases have been considered and each step verified, so the triangle inequality holds in any real inner-product space. $\square$ $\square$